

HARDWARE ENGINEERING CASE STUDY

ModDeck

Modular Adaptive Control Platform

Custom PCB design · RP2040/ESP32 firmware · pogo-pin modular
hardware architecture · KiCad schematic & layout

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Main Module + dual Mod Sub-Module stack (CAD render)

One fixed layout can't serve every workflow



The Problem

Streamers, editors, producers, and office users each need a different mix of buttons, encoders, and macros — but off-the-shelf control surfaces ship as one rigid layout.

When needs change, the whole unit gets replaced — poor cost efficiency and unnecessary e-waste.

- Streamer — scene switching, audio, chat
- Video Editor — timeline scrub, zoom, macros
- Music Producer — knobs, sliders, transport
- Office User — shortcuts, productivity macros

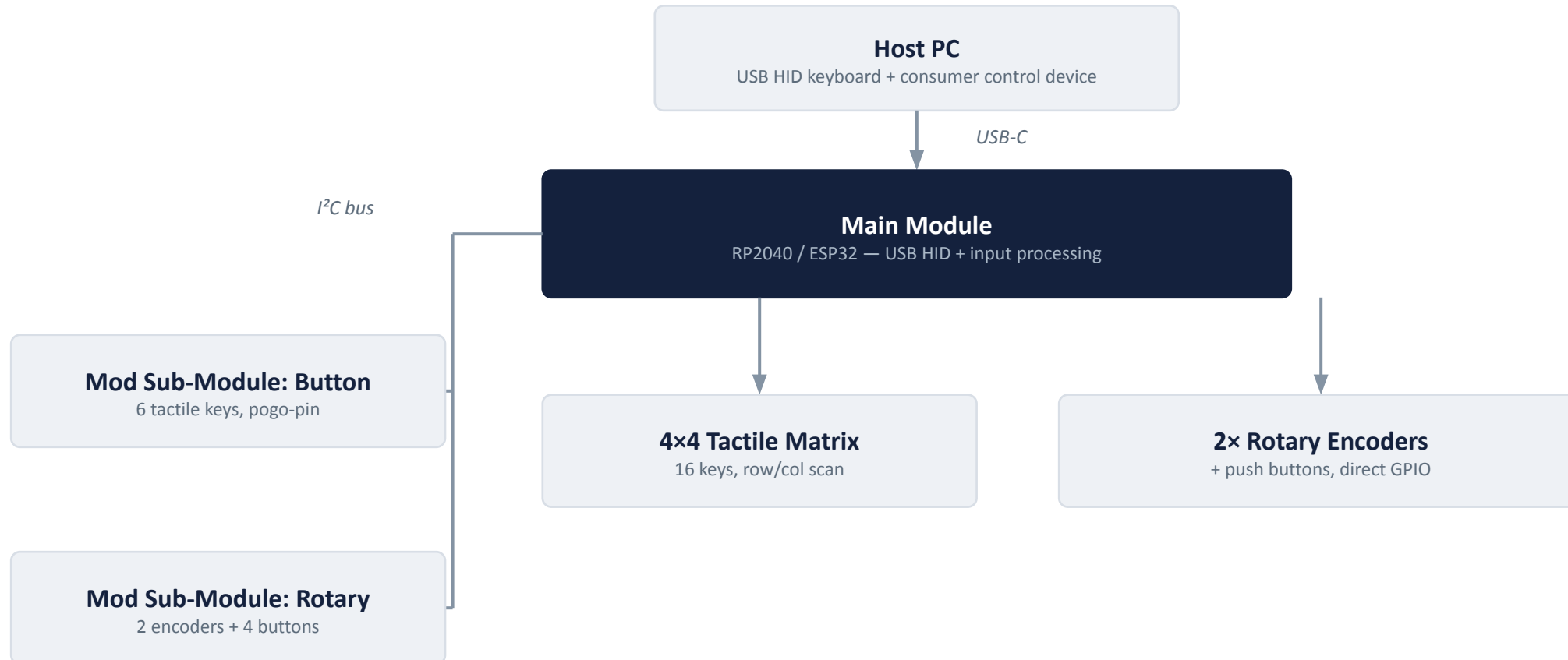


Design Vision

A **Main Module** provides the core input matrix and USB HID controller. **Mod Sub-Modules** attach via pogo-pin + dovetail rail, adding buttons or encoders on demand — reconfigured, not replaced.

Engineering goal: shift from fixed-function devices to a reconfigurable interaction system.

System Block Diagram



Communication paths: USB HID (host link, wired, zero-latency) · I²C (Main Module ↔ Mod Sub-Modules) · matrix scan (local tactile inputs).

Core Technical Architecture

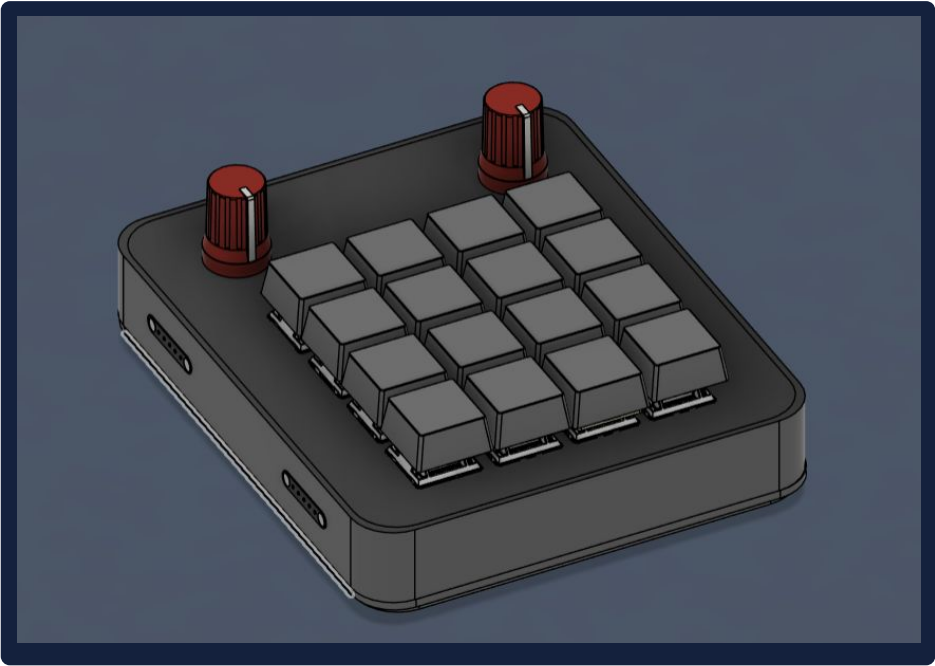
Main MCU	ESP32 or RP2040 — handles USB HID + input processing
Host communication	USB-C, wired, HID keyboard class — zero latency, no driver install
Module interconnect	Pogo pins + dovetail rail + magnetic alignment
Module protocol	I ² C bus between Main Module and Sub-Modules
Input scanning	Matrix architecture (rows × columns)

Main System Input Architecture

Core approach: matrix scanning. Instead of one MCU pin per button, ModDeck scans a rows × columns matrix — keeping GPIO count low as the key count grows.

Input type	Integration	Use case
Tactile switches	Matrix cells (4×4)	Shortcuts, macros, key presses
Rotary encoders	Dedicated GPIO ×2	Volume, scrubbing, parameters
Encoder push button	Direct GPIO	Click-to-reset, mode switch

Design reference: mechanical-keyboard matrix scanning principles, adapted for a USB HID control surface.



Expansion Module System

Purpose: add physical controls without replacing the main system.

B

Button Module

Inputs: 6 tactile keys

Connection: I²C + pogo pins

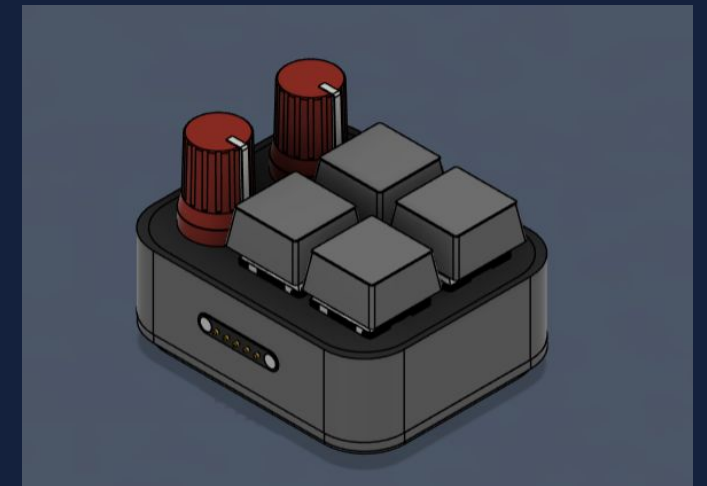
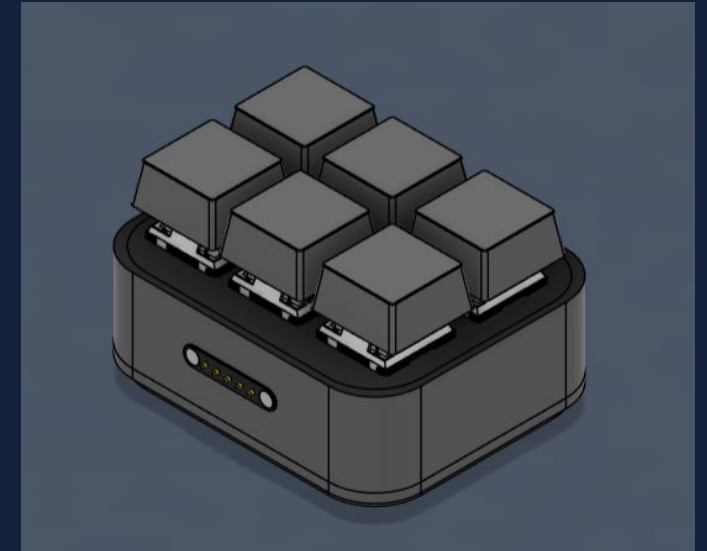
R

Rotary Module

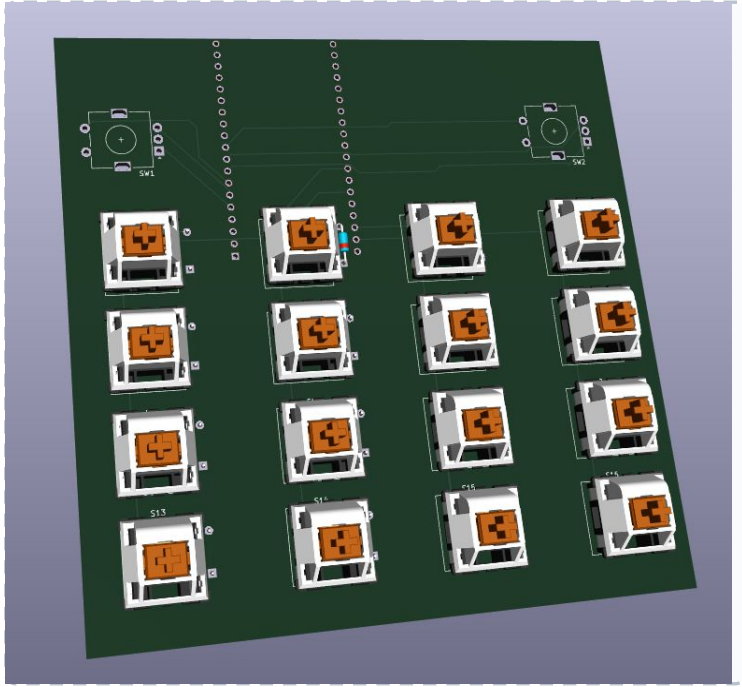
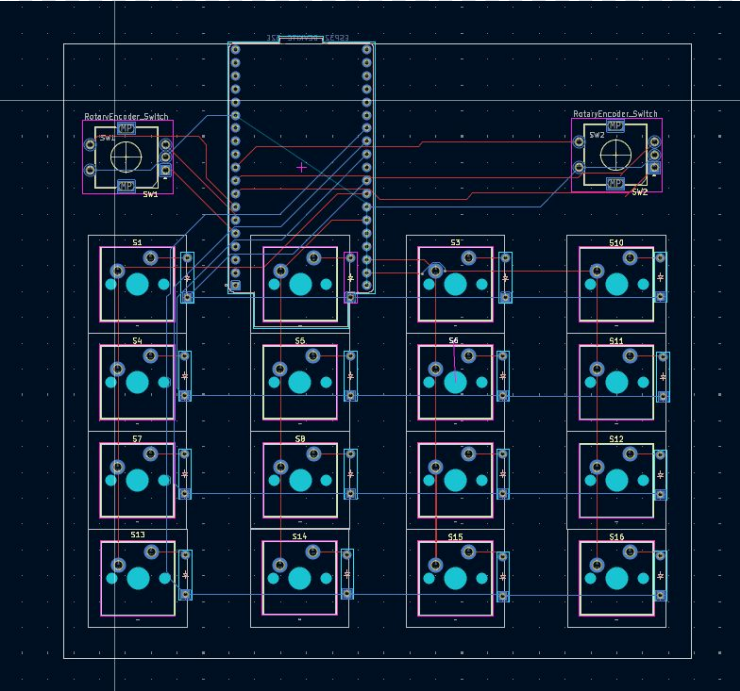
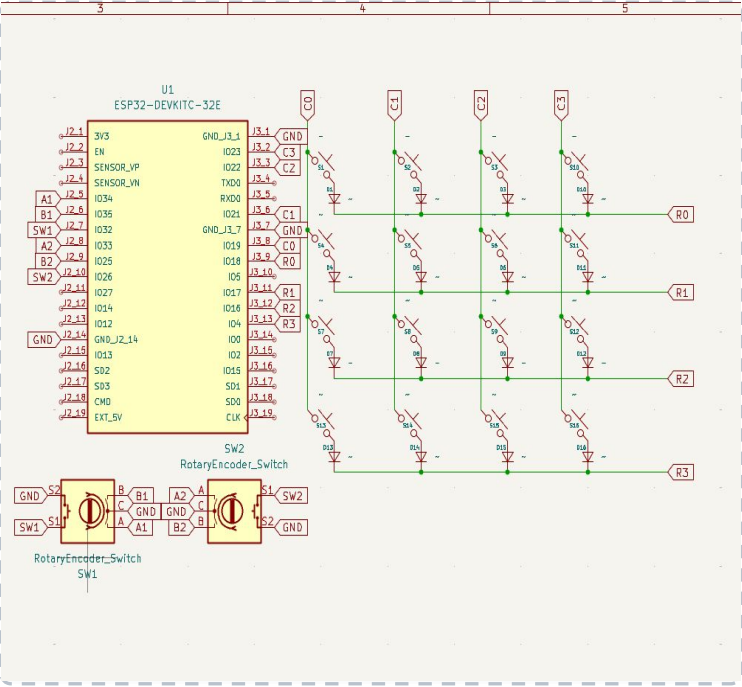
Inputs: 2 encoders + 4 push buttons

Connection: I²C + pogo pins

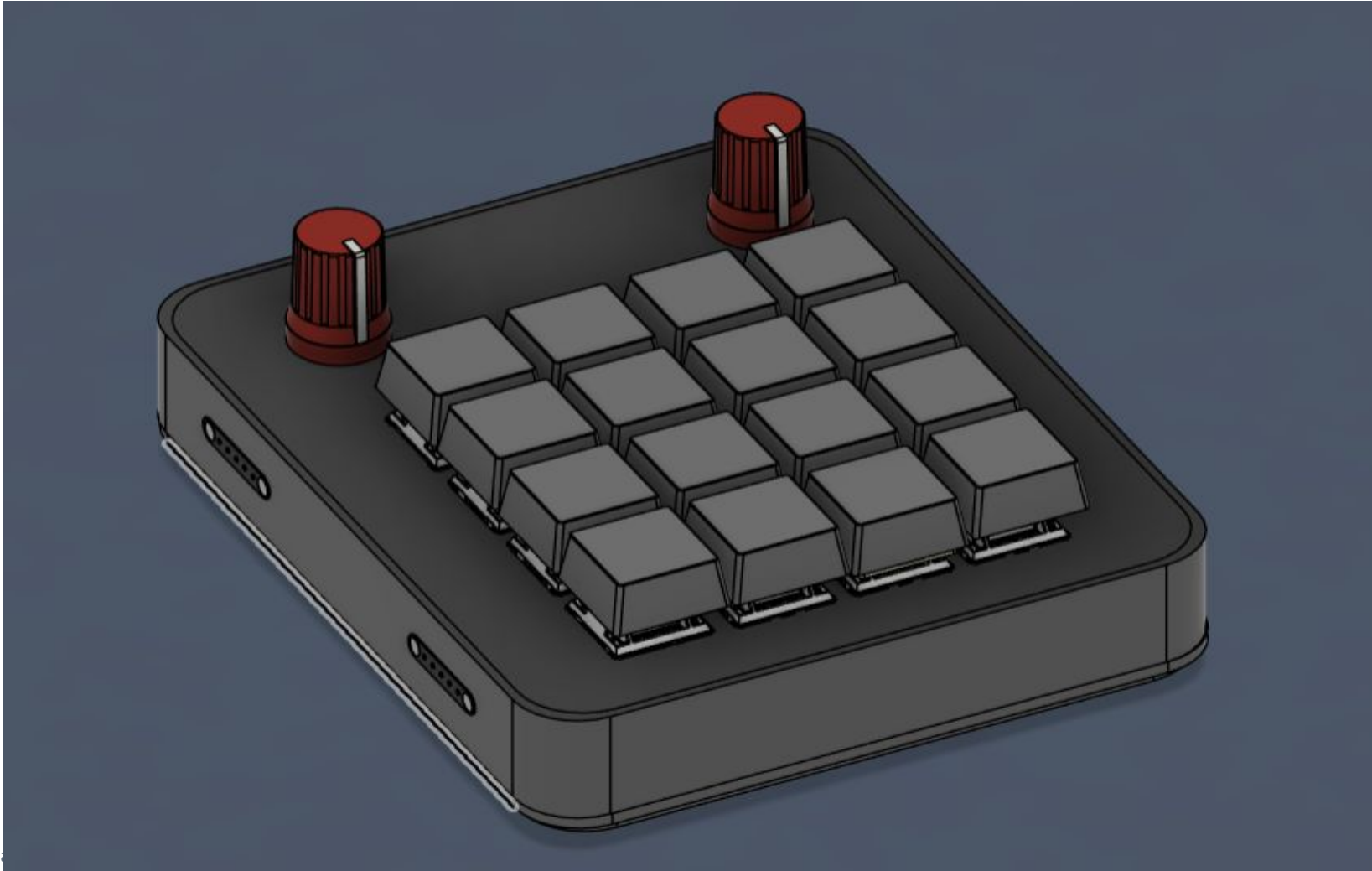
Future ecosystem: OLED display module · fader module ·
MIDI module · touch-strip module · wireless dock



KiCad Schematic & PCB Layout



Mechanical CAD — Enclosure & Assembly



Firmware & Source Code

```
matrix_scan.c · illustrative snippet

// Illustrative snippet – replace with real implementation
for (int c = 0; c < MATRIX_COLS; c++) {
    bool raw = gpio_get(MATRIX_COL_PINS[c]);

    if (raw != key_raw_last[r][c]) {
        key_raw_last[r][c] = raw;
        key_change_time[r][c] = now;
    } else if (raw != key_state[r][c] &&
               (now - key_change_time[r][c]) >=
MATRIX_DEBOUNCE_MS) {
        key_state[r][c] = raw;
        if (cb) cb((uint8_t)r, (uint8_t)c, raw);
    }
}
```



Repository

<https://github.com/AdilRehan1/ModDeck-firmware/tree/main>

README should cover:

- Build & flash instructions
- Wiring / pinout diagram
- Matrix + I²C protocol notes
- License (MIT)

Mechanical Design Decisions

Component	Design choice	Reason
Casing	Dovetail rail + magnetic alignment	Secure attachment, easy assembly
Key switches	Tactile	Affordable, reliable, satisfying feedback
Key caps	MX-compatible, ABS/PBT	Standard fit, easy replacement
Rotary encoder	24-detent incremental	Smooth rotation, infinite turn, low cost
Encoder cap	Knurled aluminum/ABS	Grip feel, professional aesthetic
Interconnect	Pogo pins, 5-pin gold-plated	10k+ mate cycles, tolerates ± 0.3 mm misalignment

Firmware & Software

System Design

- Built on RP2040 / ESP32 + USB HID
- Enumerates as a standard USB keyboard + consumer control device — no driver install
- Compatible with any key-remapping app: TouchPortal, StreamDeck-UI, MIDI mapper
- Planned: web-based configurator (WebHID) for key mapping + LED colors
- Open source (MIT) — community-extendable profiles, modules, fixes

Firmware Features

- ✓ Dynamic module detection
- ✓ Configurable key mappings
- ✓ Workflow-based profiles

Risk Register



Pogo pin oxidation / contact failure

Gold-plated tips (0.3 μm Au) + salt-spray test + user-serviceable pogo strip (\$2 part)



Mechanical misalignment

Pogo pins with ± 0.3 mm travel + dovetail with 0.05 mm clearance; tested -10°C to 50°C



ESP32 supply shortage

Footprint-compatible with STM32G0 / CH32V003 as drop-in alternates

Sustainability & Lifecycle Design

The problem: traditional control surfaces are replaced every 12-18 months, and any change in needs means swapping the whole unit.

ModDeck Solution

- Expand instead of replace
- Modular repairability
- Future-proof design

Design for Disassembly

- No glue — clips + standard screws only
- Tool-less module attachment

What Makes ModDeck Different

1

Combines four disciplines

Embedded systems + PCB engineering + mechanical modularity + HMI design in one product.

2

Adaptive physical interface

The control surface itself scales with the user's needs, not just its software.

3

A shift in device category

From fixed-function devices toward reconfigurable interaction systems.

Traction & Milestones



Hardware

- Schematic & PCB layout complete (KiCad)
- Pogo-pin module interconnect validated
- 3D enclosure designed in Fusion 360



Firmware

- USB HID working on RP2040
- Matrix key scanning implemented
- I²C module detection prototype tested



Up Next

- First PCB order (prototype v1)
- Physical enclosure print
- User testing with 5 beta users



■ MY ROLE ON MODDECK

Hardware & Firmware Lead

- Schematic capture & PCB layout in KiCad (power system, MCU, I²C module bus)
- Firmware architecture: RP2040/ESP32, USB HID, matrix scanning, module detection
- Mechanical design collaboration: enclosure, pogo-pin interconnect, Fusion 360 CAD

Tools & Stack: KiCad · Fusion 360 · C/C++ · Python · Git · Oscilloscope / Multimeter



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